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[23CSE302]

THIRD SEMESTER B. Tech EXAMINATION, DECEMBER - 2024

COMPUTER ORGANIZATION AND ARCHITECTURE

Time: 3 Hours **Maximum Marks: 100**

Instructions: i. Missing data may be suitably assumed.

ANSWER ALL QUESTIONS

		PART-A			**5 x2=10**
1.	**a.**	Define programmable logic array (PLA).	L1	CO1	**02**
	b.	Define register transfer language (RTL)?	L1	CO1	**02**
	c.	Apply binary addition to calculate the sum of 1010 and 1101.	L3	CO6	**03**
	d.	Define the role of peripheral devices in computer systems.	L1	CO3	**02**
	e.	Define memory hierarchy and its purpose.	L1	CO5	**02**
		PART-B			**6 X 5=30**
2.	**a.**	Minimize the Boolean expression F(A,B,C,D)=AB'C'D'+ABC'D'+ABC'D+ABCD'	L3	CO1	**05**
		OR			
	b.	Explain the structure and operation of a Programmable Logic Array (PLA).	L2	CO1	**05**
3.	**a.**	Describe different types of data representations in computer systems.	L2	CO2	**05**
		OR			
	b.	Discuss the working of shift micro-operations and their types.	L2	CO2	**05**
4.	**a.**	Perform binary addition and subtraction on the following numbers using 2's complement: 1101 + 1010 and 1101 - 1001.	L3	CO6	**05**
		OR			
	b.	Perform binary division on 101110 by 101 and explain each step of the division.	L3	CO6	**05**
5.	**a.**	Compare serial and parallel communication.	L4	CO3	**05**
		OR			
	b.	Explain the different modes of transfer in input/output systems.	L2	CO3	**05**
6.	**a.**	Explain the concept of parallel processing and its application in modern computers.	L2	CO5	**05**
		OR			
	b.	Discuss how pipelining improves performance in a Reduced Instruction Set Computer (RISC) architecture.	L2	CO4	**05**

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7.	a.	Compare the different types of replacement algorithms used in cache memory.	L4	CO5	05
		OR			
	b.	Discuss about cache memory design with neat diagram.	L2	CO5	05
		PART-C		**6 X 10=60**	
8.	a.	Explain the concept of bus structures in a computer system and how they facilitate communication between different components.	L2	CO1	10
		OR			
	b.	Discuss the different types of flip-flops and their applications in digital circuits with examples.	L2	CO1	10
9.	a.	Describe the entire process of register transfer language (RTL).	L2	CO2	10
		OR			
	b.	Discuss the design and functioning of a circuit that can perform all types of micro-operations	L2	CO2	10
10.	a.	Apply the binary division process to divide 11011001 by 101, and explain the remainder and quotient in detail.	L3	CO6	10
		OR			
	b.	Perform and explain binary multiplication using Booth's Algorithm for the numbers 1011 and 1101.	L3	CO6	10
11.	a.	Explain and apply Reduced Instruction Set Computer (RISC) architecture to write a program for basic arithmetic operations using a minimal set of instructions.	L2	CO3	10
		OR			
	b.	Explain briefly about various transmission modes.	L2	CO3	10
12.	a.	Describe parallel processing and its importance in solving computationally intensive tasks. Provide real-world applications.	L2	CO4	10
		OR			
	b.	Explain how instruction pipelining works in modern processors.	L2	CO4	10
13.	a.	Describe the different types of memory and explain their role in memory hierarchy.	L2	CO5	10
		OR			
	b.	Discuss the various strategies used for memory allocation and management in hierarchical memory systems.	L2	CO5	10